Exercise series 8:
3NF

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Problem 1:

A) Consider the relation schema Bookings(Title, Theater, City) with functional dependencies:

- Theater → Title
- {Title, City} → Theater

Is the original schema in *3NF* or *BCNF*?

B) If *possible,* give an example of:

1. A relation schema with dependencies that is in *3NF* but not in *BCNF*
2. A relation schema with dependencies that is in *BCNF* but not in *3NF*
3. A relation schema with dependencies that is neither in *BNCF* nor in *3NF*

Problem 2: Minimal cover

- Find a minimal cover/basis for the following relation and FDs

R(A,B,C,D,E,F,G,H)
• A → B • A,B,C,D → E • E,F → G,H • A,C,D,F → E, G

Problem 3: Decompose to 3NF

a) Which of the following tables are already in 3NF?

b) Decompose the other tables in 3NF using the algorithm

R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D)	R(A,B,C,D,E)	R(A,B,C,D,E)
• $A,B \rightarrow C$	• $B \rightarrow C,D$	• $A,B \rightarrow C$	• $A \rightarrow B$	• $A,B \rightarrow C$	• $A,B \rightarrow C$
• $C \rightarrow D$		• $B,C \rightarrow D$	• $B \rightarrow C$	• $D,E \rightarrow C$	• $C \rightarrow D$
• $D \rightarrow A$		• $C,D \rightarrow A$	• $C \rightarrow D$	• $B \rightarrow D$	• $D \rightarrow B,E$
		• $A,D \rightarrow B$	• $D \rightarrow A$		

Problem 4: Lossless and dep. preserving

Let R(A,B,C,D,E) be decomposed into relations R_1(A,B,C), R_2(B,C,D), and R_3(A,C,E).

For each of the following sets of FDs check if the decomposition preserves dependencies and/or is lossless.

- $B \rightarrow E$ and $CE \rightarrow A$.
- $AC \rightarrow E$ and $BC \rightarrow D$.
- $A \rightarrow D$, $D \rightarrow E$, and $B \rightarrow D$.
- $A \rightarrow D$, $CD \rightarrow E$, and $E \rightarrow D$.

Problem 5: Dependency preserving

Check if the following decomposition are dep. preserving.

R(A,B,C,D)

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

decomposed into $R_1(A,C)$ $R_2(C,D)$ $R_3(B,D)$

R(A,B,C,D,E,F,G)

- $A \rightarrow E$
- $B,E \rightarrow F$
- $C,F \rightarrow G$
- $A,G \rightarrow D$
- $A,B,C \rightarrow D$

decomposed into $R_1(A,D,E,G)$ $R_2(B,E,F)$ $R_3(C,F,G)$